Maxime Baelde

Ph. D, Data Scientist

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LinkedIn profile
Website

Work experiences

March 2024 – Today
Senior Data Scientist, Antvoice, Paris.
I own the machine learning behind AntVoice's real-time bidding platform and its LLM-agent products for advertisers.
- Design and own the platform's core ML models for real-time bidding, win prediction, and click/conversion forecasting, including a reinforcement-learning bidding agent
- Build and evaluate LLM multi-agent systems for advertisers: evaluation frameworks, agentic security, and automatic prompt optimization
- Industrialize data pipelines and MLOps: Airflow on GKE, BigQuery, CI/CD
- Tech: Python, GCP, BigQuery, Airflow, Docker

June 2026 – Today
Founder, Duetly, Lille.
Designed and built solo a SaaS that streamlines file transfer and revision management between audio engineers and artists, replacing WeTransfer, WhatsApp, and email.
- Audio analysis engine (LUFS, True Peak, LRA), billing infrastructure, real-time messaging, bilingual (FR/EN) interface
- File delivery, in-browser waveform playback, time-coded annotations, versioned revision cycles, quotes & invoices
- Tech: Next.js, React, TypeScript, Firebase, Stripe
- Live at duetly.app

May 2022 – February 2024
Data Scientist Consultant, Consortia, Lille.
Context: Retail
- Develop and maintain sales forecast algorithms
- Backtest and implementation of new algorithm to forecast daily store sales
- Adaptation of existing DAGs to run the tasks on Airflow
- Techno: GCP, BigQuery, GCS, AI Notebook, Airflow, Github, git

Participate in the recruitment of Data Scientists, Data Analysts and Data Engineers

October 2019 – Today
Lecturer in Statistics, Enigma School, Lille.
Lectures of Statistics with R and Python at Enigma School (Master in Computer Science, Big Data)

July 2022 – Today
Mixing/Mastering Engineer, Freelance, Lille.
Produce, record, mix and master songs for the bands The Pyloons and Fracton.

July 2019 – April 2022
Machine Learning Engineer, Data Scientist, Wavely, Villeneuve d'Ascq.
- Development of Python packages about signal processing and machine learning.
- Experiment deep learning methods for embedded audio source classification, using PyTorch.
- Conversion to ONNX format to run the algorithms on small computers
- Dockerization of audio classification service
- Update SQLite database schemas using SQLAlchemy
- Techno: Git, Python, SQL, Docker, Ansible, Grafana, Linux

November 2017 – September 2022	**Part-time lecturer in Signal Processing**, Ecole Centrale de Lille, Villeneuve d'Ascq. Involved in lectures of Signal Processing at Ecole Centrale de Lille (Bachelor and Master equivalent lectures)
October 2015 – July 2019	**R&D Engineer, Audio / Machine Learning**, Nahimic – a brand of SteelSeries, Villeneuve d'Ascq. Develop algorithms related to probabilities, statistics, machine learning and artificial intelligence, for applications to digital audio technologies. Main research topics: audio source classification, audio source separation, generative models, deep learning.
January 2016 – December 2018	**Ph. D Student**, A-Volute – Inria, Villeneuve d'Ascq. Thesis subject: Real-time identification, localization and separation of audio sources in multichannel audio streams. Supervisor: Christophe Biernacki, Professor (Université de Lille, INRIA, CNRS)
September 2018 – October 2018	**Part-time lecturer in Statistics**, ISA, Lille. Involved in lectures of Statistics with R at ISA (Bachelor equivalent lectures)
April 2015 – September 2015	**Internship – R&D Engineer**, A-Volute, Roubaix. Modelization of HRTF filters and optimization for 3D audio effect. ◦ Implementation of Machine Learning techniques (regression) for HRTF individualization ◦ Implementation of a non rigid registration algorithm for 3D scans
May 2014 – August 2014	**Internship – Research assistant**, Université Libre de Bruxelles, Brussels. Development of an experimental method for assessing the musical qualities of violin strings. Supervisor: Jean-Pierre Herman

Computer skills

Software programming	Git, GitHub, Python (Numpy, Pandas, ...), R, Bash, C/C++
Web development	TypeScript, Next.js, React.js, Firebase, Stripe
Machine Learning	Scikit-learn, PyTorch
Data Visualization	Google Data Studio, Tableau, Power BI, Grafana
Data Engineering	Google BigQuery, SQLAlchemy, SQLite, PostgreSQL
MLOps	Airflow, Docker, MLFlow, dvc, CI/CD
Audio	FFmpeg, Cubase, Wavelab Pro
Operating system	Windows, Linux

Open-source projects

churn-poc	End-to-end churn prediction project: model training to API exposition with FastAPI. [GitHub]

generative-audio-source-models	Generative models for real-time audio source classification and separation (PhD thesis, Univ. Lille 2019). [GitHub]
nongaussian-mixtures	Scikit-learn-compatible mixture models (Dirichlet, Beta, binned Gaussian) from my thesis appendices. [GitHub], [PyPI]
simple-vad	Voice Activity Detector: signal-processing beamforming + CNN classifier. [GitHub]
sober-selfhost	Energy-sober self-hosting on consumer hardware: on-demand containers (Sablier) gated behind Caddy, multi-DB encrypted backups (Postgres/MariaDB/SQLite -> rclone crypt), bidirectional Wake-on-LAN over Tailscale, and battle-tested Docker healthcheck fixes. [GitHub]

Scientific communications

2026 **Journal article (under review): TMLR**.
Maxime Baelde. "Threshold-Cascaded Lexicographic Rewards for Multi-Constrained RL in Real-Time Bidding". Submitted to TMLR (Transactions on Machine Learning Research), under review. [openreview]

2021 **Patent**.
Maxime Baelde, Laurent Mareuge and Julien Roland. "Système de reconnaissance et d'identification de sources sonores en temps réel". FR Patent FR3109458B1. [patent]

2020 **Patent**.
Maxime Baelde, Nathan Souviraà-Labastie and Raphaël Greff. "Apparatus and method for audio analysis". US Patent US11956616B2. [patent]

2019 **Ph.D thesis**.
Maxime Baelde. "Modèles génératifs pour la classification et la séparation de sources sonores en temps-réel". Ph.D thesis, Université de Lille, 2019. [thesis], [code]

2019 **Journal article: Pattern Recognition**.
Maxime Baelde, Christophe Biernacki and Raphaël Greff. "Real-Time monophonic and polyphonic audio classification from power spectra". In : Pattern Recognition 92 (august 2019), p. 82-92. [paper]

2019 **Pre-print: Gretsi 2019**.
Nathan Souviraà-Labastie, **Maxime Baelde**, Thomas Malet and Raphaël Greff. "Impact des conditions d'attaques sur les contre-mesures pour la reconnaissance du locuteur". pre-print Gretsi. [paper]

2019 **Pre-print paper: Interspeech 2019**.
Maxime Baelde, Nathan Souviraà-Labastie and Raphaël Greff. "Influence of the attack conditions on countermeasures for Automatic Speaker Verification". pre-print Interspeech. [paper]

2017 **Conference paper: ICASSP 2017**.
Maxime Baelde, Christophe Biernacki and Raphaël Greff. "A mixture model-based real-time audio sources classification method". 2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 5-9 Mar 2017, pp. 2427-2431, New Orleans, USA. [paper], [poster]

2017 **Conference paper: JDS 2017**.
Maxime Baelde, Christophe Biernacki, Raphaël Greff. "Classification de signaux audio en temps-réel par un modèle de mélanges d'histogrammes". 49èmes Journées de Statistiques, June 2017, Avignon, France. [paper]

2015 **Master Thesis**.
Maxime Baelde. “Modelization of HRTF filters and optimization for 3D audio effect”. Master thesis, 2015, Université de Lille. [paper]

2014 **Abstract: JASA 2014**.
Maxime Baelde, Jessica De Saedeleer et Jean-Pierre Hermand. “Experiment to evaluate musical qualities of violin strings”. In : The Journal of the Acoustical Society of America 136.4 (2014), p. 2284-2285. [abstract]

Education

2023 **Training**, *DJ Network*, MIXING & MASTERING (expert module): Mixing and processing audio tracks of a musical production.
RNCP35223BC04 block - Mixing and processing audio tracks of a musical production

2016–2019 **Ph.D thesis**, *Université de Lille*, Villeneuve d'Ascq, Modèles génératifs pour la classification et la séparation de sources sonores en temps réel.

2012–2015 **Engineering Degree**, *École Centrale de Lille*, Villeneuve d'Ascq, *Decision making and Data analysis*.

2014–2015 **Master Research in Applied Mathematics**, *Université de Lille*, Villeneuve d'Ascq.

2010–2012 **Preparatory classes: two-year undergraduate intensive course in mathematics and physics**, *Lycée Henri Wallon*, Valenciennes.

2007–2010 **A-Level, engineering sciences and physics speciality**, *Lycée Pierre Forest*, Maubeuge.

Language

French **Mother Tongue**

English **Technical and professional**

Hobbies

Music (technical) Compose, produce, record, mix and master songs

Music (artistic) Play the piano (since 19 years) and the guitar (since 22 years)

Videogames RPG, Tactical, Action

Reading Thriller (Maxime Chattam), science-fiction (Isaac Asimov, Alain Damasio,...)

Others Bakery, Gardening and food autonomy